Auditorne vježbe: Očitavanje signala

Zadatak 1.) Odredite FT signala:

$x(t) = \cos(100t) + \cos(200t) \qquad t \in \mathbb{R}$ → kontinuirani vremenski periodičan } CTFS

$$X_k = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-j\omega_0 k t}\, dt$$

$\omega_1 = 100 \rightarrow T_1 = \frac{2\pi}{100}$

$\omega_2 = 200 \rightarrow T_2 = \frac{\pi}{100}$

$(T_1, T_2) \Rightarrow T_0 = \frac{2\pi}{100}$

$\omega_0 = \frac{2\pi}{T_0} = 100$

$$x(t) = \frac{1}{2}e^{j100t} + \frac{1}{2}e^{-j100t} + \frac{1}{2}e^{j200t} + \frac{1}{2}e^{-j200t}$$

$$x(t) = \frac{1}{2}e^{j\omega_0 t} + \frac{1}{2}e^{-j\omega_0 t} + \frac{1}{2}e^{j2\omega_0 t} + \frac{1}{2}e^{-j2\omega_0 t}$$

$F_1 = \frac{1}{2} \qquad F_{-1} = \frac{1}{2} \qquad F_2 = \frac{1}{2} \qquad F_{-2} = \frac{1}{2}$

$F_1 \cdot 2\pi\delta(\omega - \omega_0) \qquad F_{-1} \cdot 2\pi\delta(\omega + \omega_0) \qquad F_2 \cdot 2\pi\delta(\omega - 2\omega_0) \qquad F_{-2} \cdot 2\pi\delta(\omega + 2\omega_0)$

$$\Rightarrow \boxed{X(j\omega) = 2\pi\left[F_{-2}\cdot\delta(\omega+2\omega_0) + F_{-1}\delta(\omega+\omega_0) + F_1\cdot\delta(\omega-\omega_0) + F_2\delta(\omega-2\omega_0)\right]}$$

Zadatak 2.) Odredite periode i frekvencije signalima odredenima s

$x_1(t) = \cos(6\pi t) \qquad x_2(t) = \cos(14\pi t) \qquad x_3(t) = \cos(26\pi t)$

i očitavajte ih frekvencijom očitanja $f_{s_1} = 10\,Hz$. Što primjećujete i zašto?

Što ako očitamo sa $f_{s_2} = 30\,Hz$?

$f_s > 2f$

$\omega_1 = 6\pi \longrightarrow \frac{2\pi}{\omega_1} = T_1 \;\ldots\ldots\; T_1 = \frac{1}{3}$; $f_1 = 3\,Hz$ $\qquad f_{s_1} > 2f_1 \qquad f_{s_2} > 2f_1$

$\omega_2 = 14\pi \longrightarrow \frac{2\pi}{\omega_2} = T_2 \;\ldots\ldots\; T_2 = \frac{1}{7}$; $f_2 = 7\,Hz$ $\qquad f_{s_1} < 2f_2 \qquad f_{s_2} > 2f_2$

$\omega_3 = 26\pi \longrightarrow \frac{2\pi}{\omega_3} = T_3 \;\ldots\ldots\; T_3 = \frac{1}{13}$; $f_3 = 13\,Hz$ $\qquad f_{s_1} < 2f_3 \qquad f_{s_3} > 2f_3$

nema aliasinga

aliasing!

(preklapanje spektara)

Zadatak 3.) signal $x(t) = \sin(8000\pi t) + 2\cos(24000\pi t + \frac{\pi}{3}) + \sin(16000\pi t)$

očitan je frekvencijom očitanja $f_s = 10\,kHz$. Odredite vremenski oblik signala nakon rekonstrukcije idealnom interpolatorom.

$$\omega_1 = 8000\pi \longrightarrow T_1 = \frac{2\pi}{\omega_1} = \frac{2\pi}{8000\pi} = \frac{1}{4000} = \frac{6}{24\,000}$$

$$\omega_2 = 24\,000\pi \longrightarrow T_2 = \frac{2\pi}{24000\pi} = \frac{1}{12000} = \frac{2}{24\,000}$$

$$\omega_3 = 16\,000\pi \longrightarrow T_3 = \frac{2\pi}{16000\pi} = \frac{1}{8000} = \frac{3}{24000}$$

$$\boxed{T_0 = \frac{6}{24\,000}}$$

$f_0 = 4000\,Hz$

$f_0 = 4\,kHz$

znajući da je $f_s > 2 \cdot f_0$ ne dolazi do preklapanja (aliasing)

Zadatak 4.) Za signal $x[n] = \{\ldots, 0, 0, \underset{\uparrow}{1}, 2, -1, 2, 0, 0\}$ odredite interpolacijsku fju oblika $\mathrm{sinc}(x)$. Odredite vrijednosti funkcije $t = 1.5$ uz period očitavanja $T = 1$. Podcrtani uzorak je uzorak za $n = 0$.

$\phi \quad 1 \quad 2 \quad 3$

$$x(t) = x[0] \cdot \frac{\sin(\pi t)}{\pi t} + x[1] \cdot \frac{\sin(\pi(t-1))}{\pi(t-1)} + x[2] \cdot \frac{\sin(\pi(t-2))}{\pi(t-2)} + x[3] \cdot \frac{\sin(\pi(t-3))}{\pi(t-3)}$$

$$x(t) = \frac{\sin(\pi t)}{\pi t} + 2 \cdot \frac{\sin(\pi(t-1))}{\pi(t-1)} - \frac{\sin(\pi(t-2))}{\pi(t-2)} + 2 \cdot \frac{\sin(\pi(t-3))}{\pi(t-3)}$$

$$\boxed{x(t) = \mathrm{sinc}(\cancel{\pi} t) + 2\,\mathrm{sinc}(\cancel{\pi}(t-1)) - \mathrm{sinc}(\cancel{\pi}(t-2)) + 2 \cdot \mathrm{sinc}(\cancel{\pi}(t-3))}$$

$$x(t=1.5) = \mathrm{sinc}(1.5\cancel{\pi}) + 2\,\mathrm{sinc}(0{,}5\cancel{\pi}) - \sin(-0{,}5\cancel{\pi}) + 2\,\mathrm{sinc}(-1.5\cancel{\pi})$$

$$\cancel{= \mathrm{sinc}(1.5\pi) + 2\,\mathrm{sinc}(0.5\pi) + \mathrm{sinc}(0{,}5\pi) - 2\,\mathrm{sinc}(1{,}5\pi)}$$

$$\cancel{= 3\,\mathrm{sinc}(0.5\pi) - \sin(1.5\pi)}$$

$$\left.\begin{aligned} \omega_1 &= 0.5\pi \longrightarrow T_1 = 4 \\ \omega_2 &= 1.5\pi \longrightarrow T_2 = \frac{4}{3} \end{aligned}\right\} \text{NZV} \Rightarrow T_0 = 4$$

$T_0 > 2T_s$

$4 > 2 \cdot 1$ nema aliasinga

$$\boxed{x(t=1.5) = 0}$$

Zadatak 5.) $x(t)$ je realni signal s Nyquistovom frekvencijom ω_0. Odredite Nyquistove frekvencije za sljedeće signale.

Ako signal $x(t)$ ima Nyquistove frekvencije ω_0, onda za njegovu FT vrijedi

$$X(j\omega) = 0 \quad \text{za } |\omega| > \frac{\omega_0}{2} \; ; \quad \omega_{max} = \frac{\omega_0}{2}$$

* nije frekvencija očitavanja!

$\Rightarrow$ Nyquistova f. = polovica sampling rate $\neq f_s$!

↳ najveća frekv signala koju možeš uhvatiti (npr. 500Hz ako je sampling rate 1000Hz)

→ koliko često gledaš

→ što smiješ vidjeti

mogućnost egzaktne rekonstrukcije: $f_s = \frac{1}{T} > 2 f_{max}$; $\omega_s > 2\omega_{max}$

→ Minimalna frekvencija očitavanja za koju je moguća rekonstrukcija signala x iz x_s = Nyquistova frekvencija

UVJET $\boxed{T_N < T_s}$

a) $x(t) + x(t-1) = y(t)$

pomak u vremenskoj domeni

$$y(t) = x(t) + x(t-1) \xleftrightarrow{FT} Y(j\omega) = X(j\omega) + X(j\omega) \cdot e^{-j\omega t_0}$$

ako vrijedi $Y(j\omega) = 0 \rightarrow -\cancel{X(j\omega)} = \cancel{X(j\omega)} \cdot e^{-j\omega t_0}$

$$-1 = e^{-j\omega t_0}$$

$x(t)$ je „band-limited" $\Rightarrow$ ima ograničenu širinu pojasa (sadrži frekvencije samo do $\frac{\omega_0}{2}$)

↳ ne sadrži nikakve korisne info iznad te granice

sa $x(t-1)$ ne mijenjamo frekvenciju $\Rightarrow y(t)$ ostaje unutar istog frekv pojasa

Nyquistova frekv od $y(t)$ je ω_0

b) $y(t) = \frac{dx(t)}{dt} = x'(t) \rightarrow Y(j\omega) = j\omega \cdot X(j\omega)$

↳ ne mijenjamo frekv. $\rightarrow$ ω_0 je Nyquistova frekv.

c) $y(t) = x^2(t) = x(t) \cdot x(t) = \frac{1}{2\pi}\left[X_1(j\omega) * X(j\omega)\right]$

→ kada kvadriramo signal, spektar se širi $\longrightarrow$ $\omega_n = 2\omega_0$

d) $x(t) \cdot \cos(\omega_0 t) = y(t) \xrightarrow{FT} Y(j\omega) = \frac{1}{2}\left(X(j(\omega-\omega_0)) + X(j(\omega+\omega_0))\right)$

→ množenje s kosinusom frekvencije ω_0, u spektru se signal dosljedno klonira na frekvencijama $+\omega$ i $-\omega_0$

↳ spektar više nije oko nule nego se preselio → i to 2 puta u plus i minus smjeru

▸ jedan dio spektra $\omega_0 - \frac{\omega_0}{2} = \frac{\omega_0}{2}$, $\omega_0 + \frac{\omega_0}{2} = \frac{3\omega_0}{2}$

$\downarrow$ cos, $\downarrow$ od $x(t)$, $\downarrow$ $x(t)$

▸ drugi dio spektra $-\omega_0 - \frac{\omega_0}{2} = \frac{-3\omega_0}{2}$, $-\omega_0 + \frac{\omega_0}{2} = -\frac{\omega_0}{2}$

$\Rightarrow$

zbog množenja $x(t)$ sa $\cos(\omega_0 t)$ dobivamo duplirani „pola" frekvencija

Nyquist $\downarrow$

$\omega_{max} = \frac{3\omega_0}{2} \longrightarrow \omega_s > 2\omega_{max}$

$$\boxed{\omega_s > 3\omega_0}$$

Zadatak 6.) Odredite ω_N:

a) $x(t) = 1 + \cos(2000\pi t) + \sin(4000\pi t)$

$$X(j\omega) = 2\pi\delta(\omega) + \pi\left(\delta(\omega - 2000\pi) + \delta(\omega + \omega_0)\right) - j\pi\left(\delta(\omega - 4000\pi) - \delta(\omega + 4000\pi)\right)$$

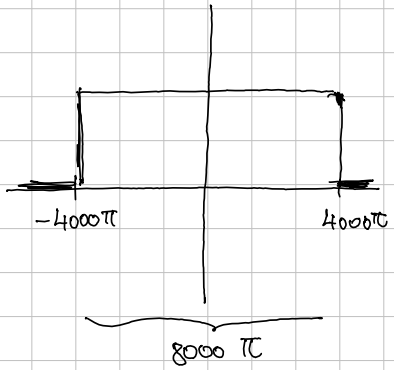
$\omega_{max} = 4000\pi \longrightarrow |\omega_N| > \omega_{max}$, $\omega_N > 2\omega_{max}$

$$\boxed{\omega_N = 8000\pi}$$

b) $x(t) = \dfrac{\sin(4000\pi t)}{\pi t} = 4000 \cdot \dfrac{\sin(4000\pi t)}{4000\pi \cdot t} = 4000 \cdot \sin(4000 \cdot t)$

$\downarrow$

$X(j\omega) = \dfrac{1}{4000}\,\mathrm{rect}\left(\dfrac{\omega}{2\pi \cdot 4000}\right) = \dfrac{1}{4000} \cdot \mathrm{rect}\left(\dfrac{\omega}{8000\pi}\right)$ pravokutna funkcija u frekv. domeni širine 8000π



znači $\omega_{max} = 4000\pi$

$\downarrow$

$\omega_N = 2\omega_{max}$

$$\boxed{\omega_N = 8000\pi}$$